Prabhavit Gupta

Forward Deployed AI Engineer: MCP Servers, Agentic Workflows & Data Foundations



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Summary

Forward Deployed AI Engineer and Lead Developer with 3 years of hands-on production experience architecting data-intensive AI systems, autonomous agent runtimes, and full-stack enterprise applications. Proven track record taking complex operational problems from whiteboard discovery to production deployment without hand-offs—building robust data models, designing custom Model Context Protocol (MCP) servers and tool plugins, and shipping Python/FastAPI microservices alongside Next.js interfaces. Deep practical specialization in applied AI pipelines: schema-enforced tool execution, deterministic validation loops, LLM-in-the-loop data transformations, and automated evaluation frameworks analyzing real production execution traces. Experienced leveraging AI coding agents as primary build accelerators to deliver hardened client deliverables in days. Holder of EU Long-Term Residence Permit (permanent unrestricted EU work authorization without visa sponsorship); based in Venice, Italy and available for immediate engagement.

Technical Expertise

- **Forward Deployed & AI Architecture:** Rapid client discovery, translating operational processes into production AI architectures, custom MCP servers, agent tools & skills, live stakeholder demos.
- **Data Engineering & Backend Stack:** Python (FastAPI, asyncio, Celery), TypeScript, Node.js, Next.js, RESTful API design, webhook synchronization pipelines, enterprise auth patterns.
- **Data Modeling & Storage Integrity:** PostgreSQL, Prisma, Supabase, relational schema design & migrations, ETL/ELT pipelines, structured JSON validation, Redis caching.
- **Applied AI & Agent Orchestration:** Multi-step agent workflows, schema-enforced tool calling, LLM grounding against live databases, guardrails, fallback escalation logic, production evaluation loops.
- **DevOps & System Ownership:** Docker containerization, AWS cloud environments, CI/CD automated deployment pipelines, Git, Playwright automated testing, logging and trace observability.
- **Languages & Authorization:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (permanent unrestricted EU work authorization, no visa sponsorship required).

Experience

FlowSpace.co — **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect — *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer support agent runtime on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Grounded agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and fallback escalation logic that routed complex inquiries to staff.
- Built automated evaluation loops analyzing real customer conversation transcripts rather than synthetic benchmarks, systematically detecting failure patterns and eliminating hallucinated answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining rapid response times.
- Deployed automated end-to-end regression test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.
- Built automated data ingestion and transformation pipelines parsing and validating unstructured partner listings, expanding platform inventory to over 900 locations.

Urlink — **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) — *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for client-facing features.
- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.

- Integrated automated CI/CD workflows and regression checks to accelerate feature delivery across distributed development environments.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.